Spatial representations emerge from temporal event credit in a hippocampus-inspired navigation agent without spatial supervision

Can spatial representations emerge from sensory inputs and learning objectives that do not explicitly encode space? We investigate this question using a bottom-up computational architecture inspired by hippocampal physiology, without position labels, maps, or spatial rewards. Rather than imposing spatial structure through an explicit learning objective, we explore whether it can emerge from the temporal organization of sensory experience and the agent's interactions with its environment.

Our model combines dentate gyrus (DG)-like sparse sensory events with CA3-like sequential memory. Temporal relationships between encountered DG events provide two complementary learning signals: a **"push"** that reinforces DG activity with more temporal separation between events, and a **"pull"** that motivates the policy to reach between internally defined DG targets sooner, making the temporal separation reflect physical distance. The agent learns from egocentric visual experience and intrinsic motivations, without explicit information about its physical location or spatial distances.

During exploration of a continuous visual environment, spatially informative DG activity emerges. To isolate the contribution of temporal information, we compare temporally weighted DG feedback against mean-matched constant feedback, evaluating the resulting representations on identical sensory observations. Temporal feedback produces higher DG spatial information, less redundant response maps, and a broader spatial distribution of field peaks across five paired training seeds. These improvements extend to downstream decoder representations, where spatial information also increases consistently. In contrast, temporally weighting the policy's target-hit reward does not consistently improve exploration or spatial representation quality compared with constant hit rewards.

The agent's behavior further shapes its learned representations. Even with network weights frozen, the learned policy produces more spatially informative downstream activity than random actions, demonstrating that spatial coding depends not only on sensory representations but also on how an agent interacts with its environment.

Together, these results show that temporally structured experience and self-generated behavior can organize hippocampus-inspired representations without explicit spatial objectives.

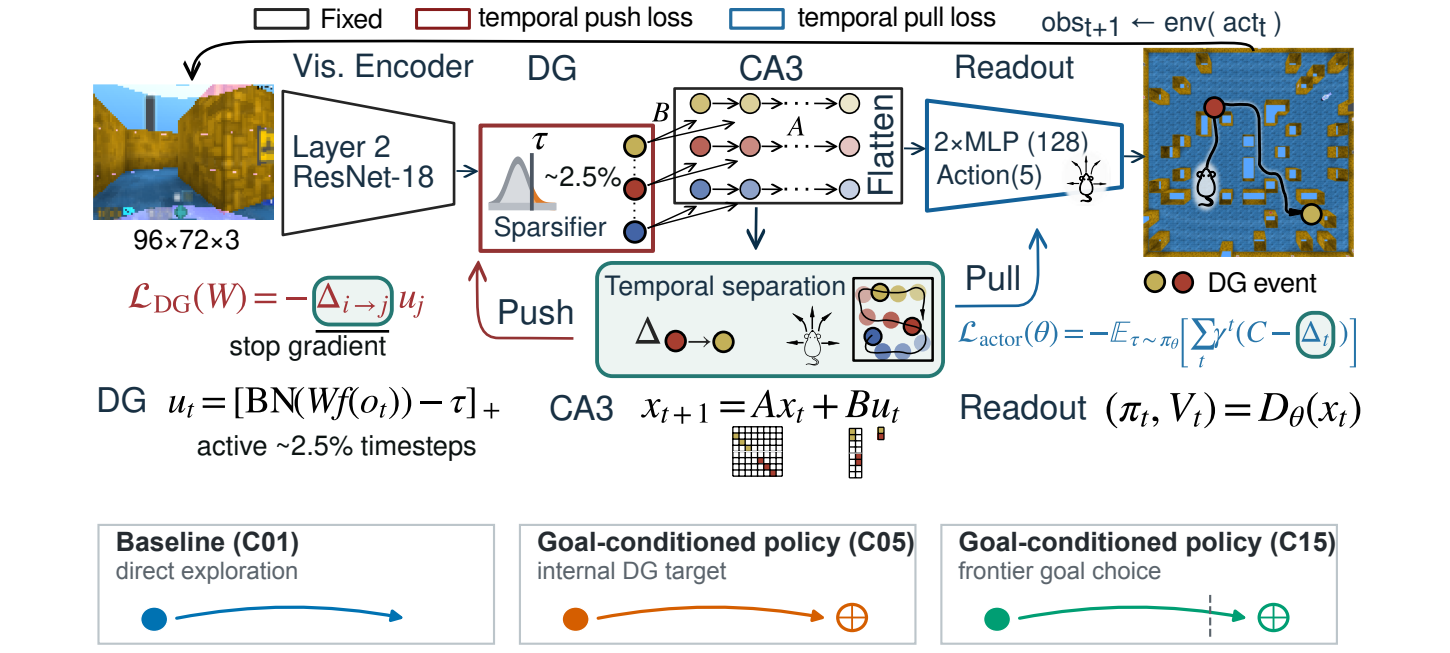


Figure 1: Push–pull architecture and variants. Fixed visual trunk, learned sparse DG, fixed CA3 sequence memory, and policy readout. Red: temporal DG credit; blue: target-hit policy credit. The strip compares baseline, goal-conditioned, and frontier-goal policies.

Spatial emergence and temporal push. DG spatial coding rises during exploration (Fig. 2A). On the same 10,001 observations, temporal DG credit increases the baseline DG spatial score in all five paired seeds (mean +0.0575), reduces map redundancy, and broadens the area covered by eligible field peaks (mean +0.109; Fig. 2C,D). Seed 8 shows the largest peak-area increase among the five pairs. Decoder-1 information also rises by 0.680 bits per activation on average in all five baseline pairs (Fig. 2E). Goal-conditioned comparisons largely point the same way; the frontier condition has only three matched encoder pairs.

Temporal pull. Replacing temporally weighted target-hit credit with constant hit credit has mixed effects on coverage (Fig. 2F) and DG fields. Both rewards still reinforce target arrival, while actor–critic discounting favors earlier arrival. Weakening PPO discounting increases frontier-policy coverage but reduces measured spatial information on separate policy rollouts. The temporal-hit panel remains a placeholder while matched-command and physical-arrival analyses are completed.

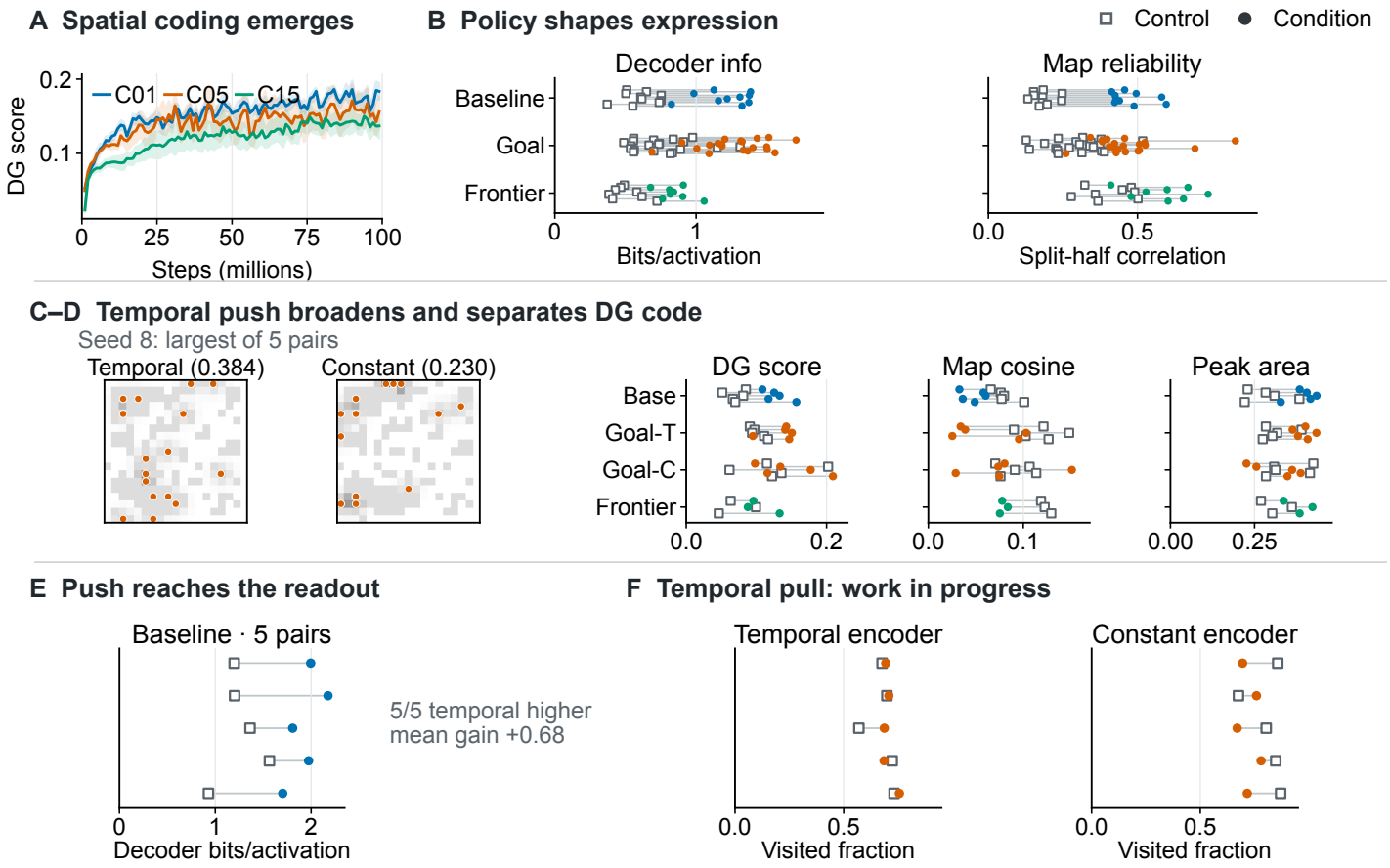


Figure 2: Results grouped by claim. Open squares show controls, filled circles show the tested condition, and gray lines join measurements from one training seed. **A,** Online DG spatial score (mean ±1 s.e.m.; five seeds per family). **B,** Random-action control versus each frozen model's own policy: decoder-1 information and split-half map reliability (38 models, two evaluation repeats averaged per model). **C,** Baseline seed 8, selected for the *largest* peak-area difference of five paired seeds: temporal 0.384, constant 0.230. Orange dots are eligible peaks after a 0.25% activity cutoff on a shared observation panel; gray bins were unvisited. **D,** Constant-credit control versus temporal DG credit for every available paired seed, showing DG spatial score, active-map cosine, and peak area on identical observations. Goal-T and Goal-C fix the temporal or constant worker bonus. **E,** Constant-credit control versus temporal DG credit in baseline decoder-1 information (5/5 higher with temporal credit). **F,** Work in progress: constant-hit control versus temporal-hit worker credit in C05 visited fraction, with the encoder fixed ($n = 5$ per arm; mean temporal-minus-constant effects +0.026 and −0.087).